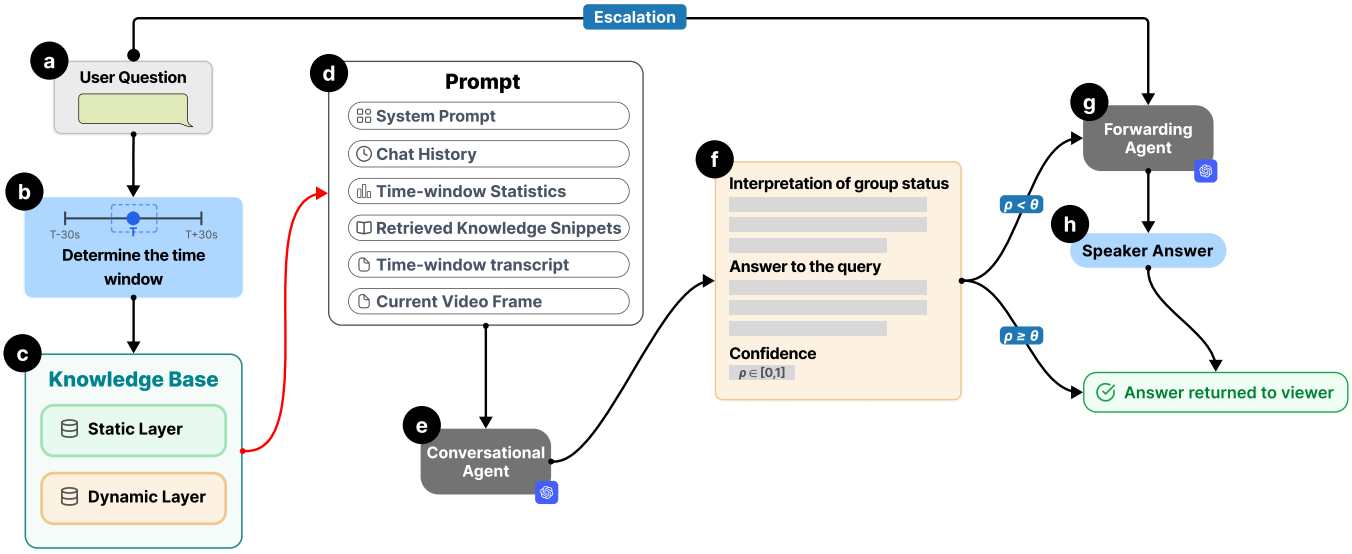


Figure 4: RAG-based Q&A Pipeline. When a viewer submits a question (a), the system determines the relevant time window centered on the current playback timestamp (b) and retrieves context from the knowledge base (c) via two parallel paths. Time-window statistics, including emoji reactions and playback behavior data, are retrieved from the dynamic layer by timestamp matching. Transcript segments within a ±30-second window around the current timestamp provide local temporal context, while vector similarity search retrieves additional relevant transcript segments, slide content, and historical Q&A records from the static and dynamic layers. The retrieved context is assembled into a structured prompt (d), together with the system prompt, chat history, and current video frame, and passed to a conversational agent (e). The agent produces a structured response (f) that includes an interpretation of group engagement status, an answer to the query, and a confidence score (ρ). If the confidence score meets the threshold ($\rho \geq \theta$), the answer is returned directly to the viewer. Otherwise, it is routed to a forwarding agent (g) for quality filtering and escalation to the speaker (h), whose expert answer is subsequently returned to the viewer. Viewers may also manually trigger escalation at any time (top).

form of pause. The behavioral signals, combined with emoji reactions and question events, aggregate into two visualizations.

The **Reaction Intensity View** (see Figure 1-b) displays the distribution of emoji reactions along the timeline as a bar chart aligned with the playback axis. The system partitions the video into adaptive temporal bins (finer for shorter videos, coarser for longer ones) and accumulates reactions within each bin. An optional kernel density estimation curve overlays the bars to show global trends. Viewers can click any bar to jump to that moment in the video.

The **Cognitive Curve** (see Figure 1-g) integrates all behavioral signals, including pauses, seeks, speed changes, emoji reactions, and questions, into a single confusion-oriented trend line. Inspired by prior work that quantifies user behavior through weighted interaction signals [59], we conducted a formative study ($N = 16$) to derive signal weights using direct relevance ratings. Participants rated the extent to which each signal reflected confusion. To account for individual differences in scale usage, ratings were normalized within participant and then aggregated across participants. The resulting values were used as empirically derived signal weights for computing the Cognitive Curve (see Table 1). The curve appears near the video player and collapses when playback controls are hidden to minimize distraction.

At peak regions of the cognitive curve, the system generates **Context Notes**: brief summaries identifying potential sources of confusion. Notes appear in a collapsible sidebar and surface only those near the current playback position, serving as lightweight guidance rather than additional reading burden.

Table 1: Behavioral signal weights. Operationalized behavior weights derived from our 16-participant formative study rating each signal's relevance to confusion. Behaviors are sorted in descending weight.

	Signal	Weight
⏪	Backward seek (rewatch)	0.190
⏸	Pause	0.172
❓	Confusion emoji	0.153
⏮	Slow down	0.150
👎	Negative emoji	0.121
⏭	Speed up	0.078
👍	Positive emoji	0.070
⏩	Forward seek	0.067

4.2.4 Escalation to Speakers. Questions escalate to speakers in two ways: automatically (when the agent's confidence falls below threshold) or manually (when viewers explicitly request human